

Benchmarking Machine Learning Against Physics-Based Proton Range Prediction Under RSP Uncertainty

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Abstract

Accurate proton range prediction is important for proton therapy, where uncertainties in tissue stopping power can affect the location of the dose distribution. This project benchmarks simple machine learning (ML) approaches against a physics-based proton transport model for range prediction under relative stopping power (RSP) uncertainty. Synthetic one-dimensional heterogeneous tissue profiles were generated using representative RSP values for lung, soft tissue and bone. Proton transport was modelled using energy-dependent stopping-power data from the NIST PSTAR database and a stepwise continuous slowing down approximation. Linear Regression and Random Forest models were trained using global features derived from the RSP profiles, including water-equivalent thickness, RSP variation, tissue fractions and initial proton energy. Performance was evaluated for 0%, 2% and 5% Gaussian RSP noise using mean absolute error (MAE), root mean squared error (RMSE) and coefficient of determination (R^2). The physics-based model remained highly accurate, with MAE increasing from 0.000 cm at 0% noise to 0.057 cm at 5% noise. In contrast, the machine learning models showed substantially larger errors of approximately 4 - 4.5 cm across all noise levels. These results suggest that the main limitation of the tested machine learning approaches was the loss of spatial information when heterogeneous RSP profiles were represented using global summary features. The study shows the importance of preserving spatial tissue information when applying machine learning to proton range prediction.

Keywords: Proton therapy; proton range; relative stopping power; machine learning; physics-based modelling; uncertainty

1 Introduction

Proton therapy is a form of radiotherapy that exploits the finite range of protons in matter and their characteristic depth-dose distribution. The resulting concentration of dose near the end of the proton path can provide advantages for delivering radiation to a tumour while reducing dose beyond the target. However, this sharp distal dose fall-off also makes proton therapy sensitive to errors in the predicted proton range. Uncertainties in range can therefore affect both target coverage and the dose received by surrounding healthy tissues [1].

A proton's range in tissue depends on the stopping power of the material through which it travels. In treatment planning, this is commonly represented by the relative stopping power (RSP), which describes the stopping power of a material relative to water. For clinical treatment planning, RSP is generally estimated from computed tomography (CT) images through a conversion between CT numbers and stopping power [2, 3]. This conversion is not exact because X-rays and protons interact differently with matter, and tissues with different compositions can have similar CT numbers but different stopping powers [3]. Consequently, uncertainty in the estimation of tissue stopping power contributes directly to uncertainty in predicted proton range [1, 4].

Machine learning methods provide an alternative approach for modelling complex relationships between physical or imaging features and clinically relevant quantities. Artificial intelligence is increasingly being investigated in proton therapy for applications including

imaging, treatment planning and quality assurance [5]. However, the use of machine learning does not necessarily provide an advantage over a model based directly on the underlying physics. For proton range predictions, a physics-based transport calculation provides a natural reference because it explicitly accounts for the energy-dependent stopping power along the proton path. A meaningful assessment of machine learning approaches should therefore consider their performance relative to such a physics-based baseline.

In this study, proton range prediction is investigated using one-dimensional heterogeneous tissue profiles with controlled RSP uncertainty. The underlying proton transport is calculated using energy-dependent stopping-power data for water from the NIST PSTAR database [6]. Machine learning models are instead provided with a small set of global features describing each RSP profile, including water-equivalent thickness, RSP variation, tissue fractions and initial proton energy. This representation provides a simple benchmark for assessing whether range information can be recovered without explicitly providing the full spatial RSP profile.

The aim of this project is therefore to evaluate simple machine learning models against a physics-based proton transport model for proton range prediction under RSP uncertainty. Linear Regression and Random Forest models are evaluated at 0%, 2% and 5% RSP noise levels using mean absolute error (MAE), root mean squared error (RMSE) and the coefficient of determination (R^2). The comparison is used to assess the effect of RSP uncertainty on prediction performance and to investigate the limita-

tions of representing heterogeneous tissue profiles using global summary features.

2 Methods

2.1 Proton Stopping Power Data

Proton stopping power data for liquid water were obtained from the NIST PSTAR database [6]. The tabulated total mass stopping power was used to calculate proton energy loss as a function of energy. Since the transport calculation requires stopping power values at arbitrary proton energies, the tabulated data were interpolated in log-energy and log-stopping power space.

2.2 Physics-Based Transport Model

A one-dimensional continuous slowing down transport model was implemented in Python. Proton energy loss was calculated according to

$$\frac{dE}{dz} = -S_{\text{water}}(E)\text{RSP}(z) \quad (1)$$

where $S_{\text{water}}(E)$ is the energy-dependent mass stopping power of water and $\text{RSP}(z)$ is the relative stopping power of the tissue at depth z . The transport equation was solved stepwise using

$$E_{i+1} = E_i - S_{\text{water}}(E_i)\text{RSP}_i\Delta z \quad (2)$$

with a spatial step of $\Delta z = 0.1$ cm. The proton range was defined as the depth at which the proton energy reached zero. The model was validated against the NIST CSDA range for protons in homogeneous water at 150 MeV [6]. Reducing the spatial step from 0.1 cm to 0.01 cm was also used to assess numerical convergence.

2.3 Synthetic Tissue Profiles and RSP Noise

Synthetic one-dimensional heterogeneous tissue profiles were generated over a 40 cm depth using representative RSP values of 0.3, 1.0 and 1.6 for lung, soft tissue and bone, respectively. Each profile contained between three and five tissue segments and was required to contain all three tissue types. Initial proton energies of 100, 125, 150, 175 and 200 MeV were considered.

For each noise condition, 1000 valid profiles were generated. RSP uncertainty was simulated by applying multiplicative Gaussian perturbations to the true RSP profile: $\text{RSP}_{\text{true}}(1 + \epsilon)$, where ϵ has zero mean and a standard deviation corresponding to the selected noise level. RSP-related range uncertainty is known to depend on the tissues traversed and on uncertainties in stopping power estimation [4]. Three noise levels were investigated: 0%, 2% and 5%. For each case, the true range was calculated using the unperturbed RSP profile. A separate physics-based prediction was calculated using the noisy profile and served as the physics baseline.

2.4 Machine Learning Models

Two regression models were evaluated: Linear Regression and Random Forest Regression. Rather than using the full spatial RSP profile, the models received a small set of global features: initial proton energy, water-equivalent thickness, RSP standard deviation, lung fraction and bone fraction. The dataset was divided into training and test sets using an 80:20 split. The physics-based range prediction was not used as an input feature to either machine learning model to avoid information leakage.

2.5 Performance Evaluation

Predicted proton ranges were compared with the true transport ranges using mean absolute error (MAE), root mean squared error (RMSE), coefficient of determination (R^2) and maximum absolute error. Performance was evaluated separately for each RSP noise level. The resulting predictions and error distributions were also examined visually using predicted versus true plots and error histograms.

3 Results

3.1 Validation of the Physics-Based Transport Model

The proton transport model was first validated using homogeneous water. For an initial proton energy of 150 MeV, the NIST PSTAR CSDA range is 15.77 cm. The model predicted a range of 16.0 cm using a spatial step of 0.1 cm, corresponding to a difference of approximately 1.5%. Reducing the spatial step to 0.01 cm gave a predicted range of 15.8 cm, demonstrating improved agreement with the NIST values and indicating convergence of the numerical transport calculation.

An example of the resulting proton transport calculation through the heterogeneous profile is shown in Figure 1. The true and perturbed RSP profiles illustrate the spatial variation introduced by the 5% noise model (Figure 1). The corresponding proton energy decreases with depth until the proton stops, demonstrating the stepwise transport calculation (Figure 1).

3.2 Machine Learning and Physics-Based Performance

The performance of the physics-based baseline, Linear Regression and Random Forest models is summarised in Table 1. The physics-based model remained highly accurate as RSP noise increased. Its MAE increased from 0.000 cm at 0% noise to 0.057 cm at 5%, while its R^2 remained approximately 1. In contrast, the machine learning models showed substantially larger errors, with MAEs of approximately 4.1 - 4.5 cm across the tested noise levels. Figure 2 shows that the physics-based error increased slightly with RSP noise, while the machine learning errors remained around 4-4.5 cm without a clear monotonic dependence on noise level.

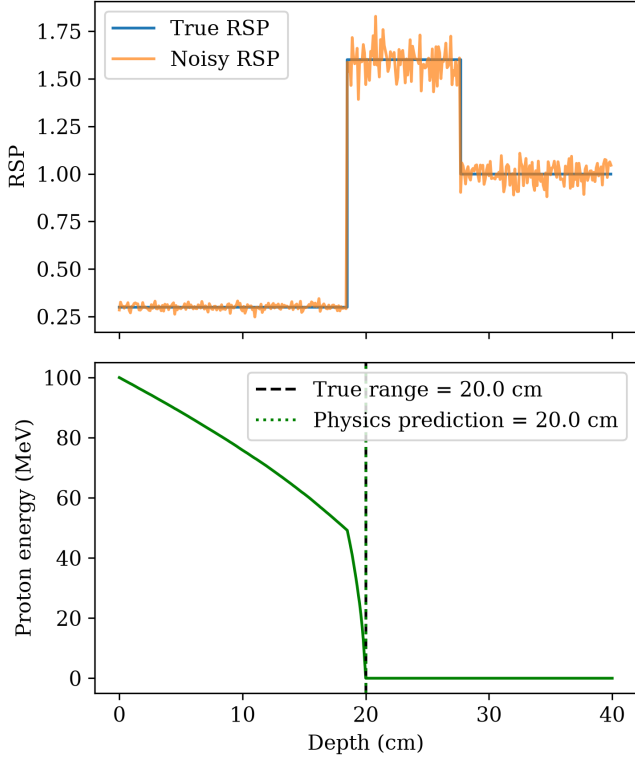


Figure 1: Example physics-based proton transport calculation for a heterogeneous tissue profile at 5% RSP noise. At the top: True and noisy RSP profiles. At the bottom: Proton energy as a function of depth up to the stopping point through heterogeneous tissue (100 MeV).

At 5% RSP noise, the predicted versus true range plot for Linear Regression shows a broad distribution around the ideal ($y = x$) relationship (Figure 3). The model captured the overall relationship between the input features and proton range, but individual predictions showed substantial deviations from the true values.

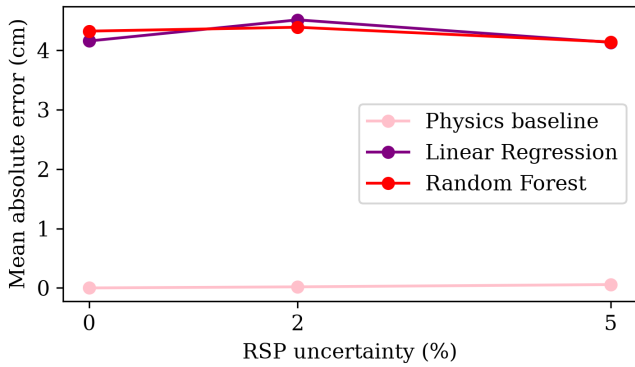


Figure 2: Model performance versus RSP noise level. Mean absolute error (MAE) for the physics-based baseline, Linear Regression and Random Forest models at 0%, 2% and 5% RSP noise.

3.3 Distribution of the Prediction Errors

The prediction error distributions for the two machine learning models at 5% RSP noise are shown in Figure 4. Both distributions were broadly centred around zero but

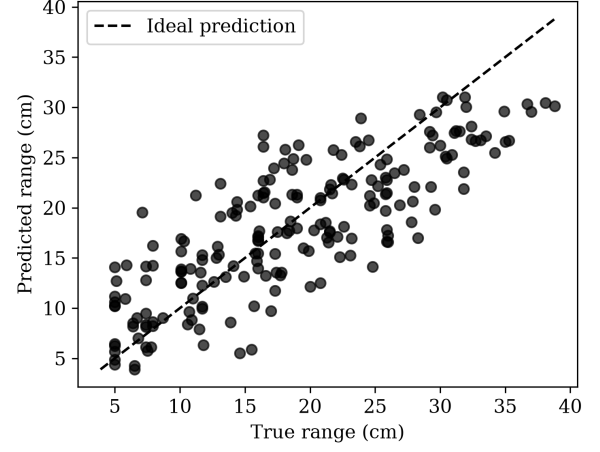


Figure 3: Predicted versus true proton range for the Linear Regression model at 5% RSP noise. The dashed line represents ideal agreement.

showed extended tails, indicating substantial errors for some heterogeneous profiles. The maximum absolute errors were 12.42 cm for Linear Regression and 17.99 cm for Random Forest. Overall, the physics-based model provided substantially more accurate range predictions than either machine learning model under all tested conditions.

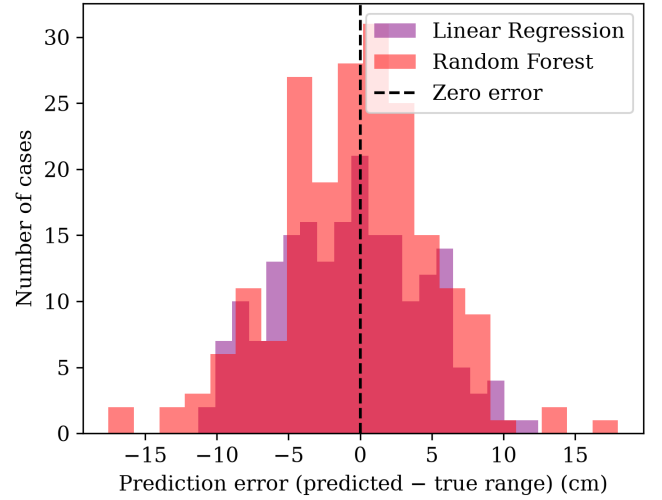


Figure 4: Distribution of proton range prediction error for Linear Regression and Random Forest at 5%. Prediction error is defined as predicted range minus true range.

Table 1: Model performance versus RSP noise level.

Noise level (%)	Model	MAE (cm)	RMSE (cm)	R^2	Max error (cm)
0	Physics baseline	0.000	0.000	1.000	0.000
	Linear Regression	4.155	5.149	0.625	12.135
	Random Forest	4.321	5.485	0.574	19.701
2	Physics baseline	0.019	0.045	1.000	0.200
	Linear Regression	4.511	5.451	0.665	13.065
	Random Forest	4.387	5.793	0.622	20.927
5	Physics baseline	0.057	0.095	1.000	0.500
	Linear Regression	4.132	5.052	0.656	12.420
	Random Forest	4.139	5.368	0.611	17.987

4 Discussion

The results show a clear difference between the physics-based and machine learning approaches to proton range prediction. The physics-based model remained highly accurate when RSP uncertainty was introduced, with the MAE increasing from 0.000 cm at 0% noise to only 0.057 cm at 5% noise. In comparison, the Linear Regression and Random Forest models had MAEs of approximately 4–4.5 cm across all noise levels. This demonstrates that, for the simplified problem considered here, explicitly modelling the proton energy loss provides substantially more accurate range predictions than the tested machine learning approaches.

The relatively large machine learning errors were already present in the absence of RSP noise. This suggests that the main limitation was not the uncertainty model itself, but the representation of the heterogeneous tissue profiles. The ML models used global summary features such as water-equivalent thickness, RSP variation and tissue fractions. These features describe the overall composition of the material but do not retain information about the spatial ordering of tissues. Two profiles can therefore have similar global features while producing different proton ranges because the same tissues occur at different depths. The physics-based model, in contrast, uses the complete spatial RSP profile during transport and therefore retains this information.

The machine learning performance also did not show a clear monotonic dependence on noise level. The errors remained within a similar range between 0%, 2% and 5% noise, while the physics-based model showed a gradual increase in error. This indicates that increasing RSP noise was not the dominant source of error for the ML models within the tested range. The similar performance of Linear Regression and Random Forest further suggests that increasing model complexity alone was not sufficient to overcome the information loss caused by the global feature representation.

These findings highlight an important consideration when applying machine learning to proton therapy. Machine learning models are being explored across several proton therapy applications, but their clinical usefulness depends on appropriate input representations, validation and comparison with established physical methods [5]. In this study, the physics-based approach provides a strong

reference against which future, more spatially aware ML methods can be evaluated.

5 Limitations and Future Work

This study has several limitations. The tissue profiles were synthetically generated using simplified one-dimensional RSP distributions and representative values for lung, soft tissue and bone. The proton transport model also used a continuous-slowness-down approximation with stopping power data for liquid water scaled by RSP, rather than a full Monte Carlo treatment. In addition, the machine learning models were trained using a small set of global summary features, which limits their ability to represent spatial variations in tissue composition. The results should therefore be interpreted as a controlled benchmark rather than a clinically realistic assessment of proton range prediction.

Future work could address these limitations by using patient-derived CT or RSP distributions and models that directly process the spatial profile. One-dimensional convolutional neural networks or other profile-aware architectures could preserve the positional information that was lost in the present feature-based approach. Another possible direction would be residual learning, in which a machine learning model predicts a correction to the physics-based range rather than the range directly. More realistic uncertainty models and validation against Monte Carlo simulations could also provide a stronger assessment of range uncertainty in clinically relevant geometries [1].

6 Conclusion

This study benchmarked Linear Regression and Random Forest models against a physics-based proton transport model for range prediction under RSP uncertainty. The physics-based approach remained highly accurate up to 5% RSP noise, while the machine learning models showed substantially larger errors across all noise levels. The results indicate that the main limitation of the tested ML approaches was the loss of spatial information when heterogeneous RSP profiles were reduced to global summary features. Overall, the study demonstrates the value of physics-based modelling as a benchmark for machine learning approaches and highlights the importance of

preserving spatial tissue information in future ML-based proton range prediction methods.

Data and Code Availability

The proton stopping-power data used in this study were obtained from the NIST PSTAR database. The synthetic RSP profiles and analysis code were developed in Python for this project. Source code, data-generation scripts and analysis notebooks are available in the accompanying GitHub repository.

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